

Executive Summary

TokenHarvest will implement a **member-driven tea trading platform** combining an **English (open, ascending) auction engine** and a complementary **direct-listing marketplace**. It uses electronic Warehouse Receipts (eWRs) to guarantee each lot's quantity and quality, enabling secure trading and instant financing. The design follows international standards (Mombasa, Colombo auctions and tea board rules) and East African commodity regulations, while adding modern auction rules (tiered tick sizes, anti-sniping soft close, mandatory bid deposits) and broker/warehouse agent workflows. A typical lot catalog includes all traceability and quality fields (grade, weight, origin, GI, certifications, etc.), plus auction parameters (reserve, increments, start/end times). All buyers and brokers are KYC/AML-verified, with hard bid-security and *"Prompt Date"* payment deadlines. Post-auction, the system automates settlement, ownership transfer of the eWR, fee payments, invoicing and delivery orders.

This document details functional requirements, data models, workflows, UI mockups, APIs, security/compliance rules, and reporting for the **TokenHarvest Tea Auction Engine and Marketplace**. Key features include:

- **Roles & Permissions:** Producers, Brokers (selling agents), Warehouses, Buyers, Financiers and Platform admins (KYC, onboarding, trading limits). Brokers list on behalf of owners under digital mandates.
- **Catalogue Types:** *With valuation* (broker attaches target price) or *Without valuation* (pure description) – see **Table 1**.
- **Auction Rules:** Open ascending auction per lot, confidential reserve, tiered bid increments (e.g. $\geq 0.5\text{--}1.5\%$), 3-minute anti-sniping extensions (max ± 30 min), hidden bidder identities. Sealed bids and reverse auctions also supported where needed. Reserve failures allow brokers to withdraw ("take out") lots **[6†L312-L320]** **[6†L323-L331]** .
- **Listing Engine (Direct Sales):** Sellers or brokers can post fixed-price lots (with negotiation or *"Make Offer"*), subject to owner's approval and usual data fields. Payment and delivery handled via the platform.
- **Pre/Post-Auction Docs:** Digital dispatch forms (pre-auction dispatch instructions from factory to warehouse), warehouse arrival/weight reports, Delivery Orders, invoices, commission statements, and compliance reports are auto-generated.
- **Settlement & Compliance:** The buyer pays by the *Prompt Date* (default 10 business days). The system holds funds in escrow, deducts marketplace and broker fees (typically 1.0% sellers, 0.5% buyers **[46†L92-L94]**), then transfers net proceeds to the owner. Only after cleared payment does a Delivery Order release goods **[6†L381-L389]** . Defaults incur penalties (funds forfeiture, future bidding bans). All actions are audited.
- **Data & APIs:** A comprehensive data schema captures every lot's info (see **Table 2**). RESTful APIs allow catalogue upload, lot search, bidding, payment instructions, etc. Sample JSON structures are provided below.
- **User Interface:** Simple, mobile-responsive screens for catalogue entry, broker portals, live auction dashboards, bid entry, dispatch management, and settlement reports. (Sample wireframes shown.)

Together, these components create a robust “Tea Exchange and Marketplace” that brings digital transparency, fairness and liquidity to the Kenyan/Sri Lankan/Indian tea trade, aligned with ICC/ISO and East African market standards.

Roles & Permissions

- **Producers/Owners:** Teafactories, cooperatives or traders owning the tea/eWR. They supply product, quality data, and appoint brokers/agents. They cannot bid.
- **Brokers (Selling Agents):** Licensed (via EATTA/CTTA rules) to act *exclusively* for one or more owners. They *must not* have conflicts (e.g., can’t own production). Brokers prepare catalogs, advise on quality/pricing, and manage sales [4†L178-L185] [46†L92-L94] . They earn commissions (typically 1% from sellers, 0.5% from buyers [46†L92-L94]) on each lot sold. The platform records formal *agency mandates* (authority, commission, term) between owner and broker. Brokers can list and close sales for their principals but **cannot** transfer funds out of the owner’s account.
- **Warehouse Operators:** Licensed warehouses issue and update eWRs and store tea. They provide gross/net weights, inspect condition on receipt, and handle dispatch. They may act as *listing agents* if authorized by the owner (e.g. farmer deposits and authorizes sale by warehouse). The platform tracks storage fees and warehouse identity for each lot.
- **Buyers:** Registered institutional purchasers (importers, blenders, exporters, traders). They bid in auctions or buy listings. Must pass KYC/AML and have sufficient approved credit/wallet balance. Buyer identities remain anonymous during auctions to ensure fair competition.
- **Financiers:** Banks or SACCOs connected via API. They can pre-approve buyer credit lines or purchase auction-winning eWRs and provide immediately funds to sellers. The system flags financed lots and automates lender notifications.
- **Platform Admins (TokenHarvest):** Onboard/verify members, set trading limits, configure fee rates, and ensure compliance with the WRS Act and auction rules.

All user actions are role-restricted and audited for compliance. Brokers and warehouses are independently certified and listed under WRS regulations [25†L251-L259] [46†L92-L94] .

Auction Engine: Functional Workflow

The Auction Engine drives *weekly scheduled auctions* (e.g. Tea: Mond/Tues, as per market). Each auction consists of multiple lots offered in a predefined order (catalogue). The workflow is:

1. **Catalog Creation:**
2. The broker logs in and prepares the **auction catalogue**, assigning each factory invoice a unique Lot ID. For each lot, the broker enters all data fields (see **Table 2**), attaches quality certificates, and sets a confidential *reserve price* (typically at or above their valuation). (This follows the Mombasa/Colombo practice of broker cataloguing and valuation [4†L178-L185] [4†L250-L256] .)
3. The platform validates entries (e.g. checks weights match warehouse records, eWR ownership).
4. **Catalog Type:** Optionally, brokers attach a “*broker valuation*” (numeric), making it a **with-valuation** catalogue. Otherwise it’s **without-valuation** (pure specs, as some markets require). See **Table 1**.

5. Sample Distribution (Tea-specific):

6. For each lot, the broker has already deposited a 4 kg sample with each buyer (or uploaded sample info) 【4†L222-L227】. The system confirms sample receipt by buyers (or handles proxy tasting comments).

7. Auction Parameters:

8. Auction *start/end time* is set by the system (e.g. Tuesday 8:30–12:00). The auction clock is monitored server-side.
9. **Tick Schedule:** Bids must exceed the previous high bid by at least the greater of a fixed tick or a percentage. We use tiered increments: e.g. 1.5% for low-value lots, 1.0% for mid-range, 0.5% (or fixed KES 500) for high-value auctions. (Historically, Mombasa used ₹0.01/kg steps 【34†L58-L62】; TokenHarvest’s approach avoids giant jumps on big trades.)
10. **Sniping Control:** If any bid arrives within the last 3 *minutes* of the scheduled close, the auction is extended by +3 minutes. This “soft-close” anti-sniping rule allows responses to last-second bids. A cap (e.g. max +30 min) prevents runaway extensions.

11. Live Bidding:

12. When live, the current lot’s details (quantity, grade, warehouse, etc.) and the latest bid are shown to all buyers. Bidders submit **binding** bids (in USD or local currency). They must have approved funds/ credit for the bid (lock on wallet).
13. Each new bid $\geq$ (current_high + Δ), where Δ follows the tick rule above. Buyer IDs are hidden to all except their own view.
14. The auctioneer (automated) locks in the highest bid if time expires with no new bid. If the final bid $\geq$ reserve, **the lot is sold** (“knocked down”) to that buyer 【6†L312-L317】. If not, the broker may “take out” the lot, withdrawing it from the sale 【6†L319-L328】.
15. **Special Cases:** Multi-pallet lots can be *split* (“shared”) if multiple buyers agree at the final price 【6†L333-L342】 (TokenHarvest will allow lot division per agreed share). The system supports multi-unit auction logic for this.
16. Sealed-bid mode and reverse-auction mode can be configured separately (e.g. government tenders or procurements).

17. Post-Auction Processing:

18. Immediately upon sale, the system generates a **Confirmation of Sale** and holds the buyer’s payment in escrow (debiting their wallet or instructing bank transfer). It also creates invoices and trade records for buyer and seller, noting lot details, prices, taxes, and fees.
19. The **Prompt Date** countdown (e.g. 10 working days) is set per EATTA rules 【6†L364-L372】. If payment is not confirmed by then, penalties apply (part-forfeiture, registration of lien, barring from future auctions 【6†L364-L372】).

20. Once payment is received and verified (the iTTS approach is to integrate with payment systems for automatic clearing **【6†L370-L374】**), the system notifies the broker. The broker issues a **Delivery Order** to the warehouse, after which the buyer may arrange pickup **【6†L381-L389】** .

21. **Settlement:**

22. Upon payment, the platform's settlement engine automatically allocates funds: deduct marketplace fee, broker commission (e.g. 1% and 0.5% as per **【46†L92-L94】**), and any financing fees if a lender financed the purchase. Net proceeds are credited to the owner.

23. The **eWR ownership** is transferred to the buyer in the central registry, effectively transferring title. The lot remains in warehouse until the buyer withdraws it (or resells the eWR).

24. The warehouse updates the registered owner of record. Storage costs (paid or unpaid) are updated; the buyer assumes any future storage.

【58†embed_image】 *Figure: Conceptual illustration of warehouse receipts and collateralized inventory in a structured marketplace. TokenHarvest's e-auction ensures that stored tea can be safely used as tradeable collateral, unlocking finance and price discovery **【25†L251-L259】** .*

Catalogue Data Model

Each auction **Lot** (or listing) is a record with the following fields. **Table 2** below lists all fields in a catalog entry (drawn from tea auction and commodity finance best practices). These feed both the Auction and direct listing engines. Some fields apply only to auctions (reserve price, lot division), while others (e.g. auto-bid) may be global.

Field	Description	Example/Data Type
Lot ID	Unique identifier for the lot/ listing (platform-generated)	"LOT-20230715-001" (string)
eWR Number	Electronic Warehouse Receipt ID (if warehoused)	"EWR-100045"
Owner ID	Beneficial owner (producer's account ID)	"USER-1002"
Broker ID	Listing agent (licensed broker/ who made listing)	"BROKER-12"
Warehouse ID	Storage warehouse code/ID	"WH-ELD-02"
Warehouse Location	Region/city of warehouse	"Eldoret, Kenya"
Invoice #	Factory invoice number	"INV-20230701"
Grade	Tea grade name/class	"OP1 Dust CTC"

Field	Description	Example/Data Type
Grade Mark	Grade mark or code from factory/tea board	"Lunda"
Quantity (Gross)	Total weight including packaging (kg)	100000
Quantity (Net)	Net weight of tea only (kg)	98100
Tare Weight	Weight of packaging/pallets (kg)	1900
Package Type	E.g. Paper sacks, Cartons, Cartons (100g pkts)	"Paper sacks"
Packing Weight	Net weight per package or pallet (kg)	50
Traceability/GI	Geographic Indication or origin trace (e.g. region)	"Nyeri-ABK"
Taster Remarks	Short notes from quality assessment	"Golden liquor, robust"
Certifications	Organic, Fairtrade, Rainforest (list)	["Organic", "Rainforest Alliance"]
Storage Status	Current storage info (days stored, cost rate, etc.)	{"days": 30, "rate": 5}
Financing Status	Flags if lot is pledged or financed (none/pledged)	"None" or {"pledge": true, "bank": "XYZ"}
Listing Type	"Auction" or "FixedPrice"	"Auction"
Reserve Price	Confidential floor price (per kg) for auction	\$2.40/kg
Valuation (if any)	Broker's estimated value (per kg, optional)	\$2.60/kg
Commission Rate	% payable to broker on sale (optional override)	0.0125 (1.25%)
Start Time	Auction start datetime (UTC)	2026-07-15T07:00:00Z
End Time (scheduled)	Auction scheduled end datetime	2026-07-15T11:00:00Z
Tick Size Schedule	Structure for bid increments (tiers or fixed)	[{"max": 500000, "inc": "1.5%"}, {"min": 500000, "inc": "1.0%"}]

Field	Description	Example/Data Type
Anti-Snipe Rules	e.g. extend by 3 min if bid in last 3 min (max ext)	<code>{"extend":180, "limit":1800}</code>
Bid Security Required	% of expected trade value held (0 if none)	<code>0.02</code> (2%)
Auto-Bid Enabled	Allow buy-it-now or proxy bidding flag	<code>true</code>
Visibility	<code>"Public"</code> or <code>"Private"</code> (who can see listing)	<code>"Public"</code>

Table 1: Pre-auction catalogue types. With valuation includes broker's price guidance and is typical in structured markets [4†L250-L256] ; without valuation omits it (common in raw commodity postings).

Table 2: Tea Lot Listing Schema. Each auction lot or listing record includes full traceability (grade, GI, certificates), storage/financing state, and auction parameters (reserve, tick rules, etc.). The `Tick Size` `Schedule` and `Anti-Snipe` fields encode the auction's incremental and extension rules (see below).

```
// Sample JSON for a lot listing (Auction type with valuation)
{
  "lotId": "LOT-20230715-001",
  "eWR": "EWR-100045",
  "ownerId": "USER-1002",
  "brokerId": "BROKER-12",
  "warehouse": {"id":"WH-ELD-02","location":"Eldoret, KE"},
  "invoice": "INV-20230701",
  "grade": "OP1 Dust CTC",
  "gradeMark": "Lunda",
  "originGI": "Nyeri-ABK",
  "quantityKg": 98100,
  "tareKg": 1900,
  "packageType": "Paper sacks",
  "packingWeightKg": 50,
  "tasterRemarks": "Sweet aroma, brisk",
  "certifications": ["Organic"],
  "storageDays": 30,
  "financing": {"status":"None"},
  "listingType": "Auction",
  "reservePrice": 2.40,
  "valuation": 2.60,
  "commissionRate": 0.01,
  "auctionParams": {
    "startTime": "2026-07-15T07:00:00Z",
    "scheduledEnd": "2026-07-15T11:00:00Z",
    "tickTiers": [
```

```

        {"upTo": 500000, "incrementPct": 1.5},
        {"above": 500000, "incrementPct": 1.0}
    ],
    "antiSnipe": {"extendSec": 180, "maxExtendSec": 1800}
},
"bidSecurityPct": 2.0,
"autoBid": false,
"visibility": "Public"
}

```

Auction Rules & Business Logic

- **Auction Type:** Open ascending (English) auction by default, as in Colombo/Mombasa practice [46†L92-L94] [6†L312-L320] . Sellers may specify “forward contract” delivery dates for seasonal teas, which the platform treats as normal lots with later delivery obligations.
- **Tick Size:** Each new bid must exceed the current bid by at least the tiered increment. For example, for lots < KES 500,000 use 1.5% increments; for larger lots 1.0% or 0.5% (or fixed KES tick). This prevents bid jumps from stalling smaller trades. (By comparison, Mombasa auctions historically use ₹0.01/kg steps [34†L58-L62] .)
- **Anti-Sniping:** If a bid arrives within N minutes of scheduled close, the close is extended by N minutes. We set $N=3 \text{ min}$, with a total extension cap (e.g. 30 min) to ensure certainty. This *soft-close* avoids unfair “last-second” wins.
- **Sealed Bid / Reverse Auction:** The system can run sealed bids (highest bid wins but not visible) or reverse auctions (buyers post RFQs, sellers bid down prices) for special cases.
- **Division/Sharing:** Lots are typically whole (e.g. full pallets). The system allows *lot-splitting* at the final price if requested by smaller buyers, mirroring Mombasa’s lot-sharing [6†L333-L342] .
- **Bid Security:** To deter bogus bids, each bidder must have funds (or escrow) for a percentage of lot value (e.g. 1–5%). Winning bidders failing to pay forfeit their deposit.

Broker Mandates & Authorization: Each broker listing a lot must hold a signed digital mandate from the owner specifying permissions (e.g. list, accept bids, negotiate price, commission rate, validity period). Owners can change or revoke mandates at any time; the system will enforce that only the authorized broker can manage a given lot. Multiple brokers **cannot** list the same lot (the system locks it to one agent) unless the owner explicitly allows co-listing (a rare case).

Prompt Date & Payment: The *Prompt Date* (final payment deadline) is 10 working days from sale [6†L364-L372] , as per EATTA rules. The system will **not** release the Delivery Order until full payment clears. Late or defaulting buyers face penalties (e.g. interest on delayed payment, loss of deposits, temporary suspension). This strict payment discipline aligns with global tea auction norms [6†L364-L372] .

Settlement & Penalties: On closing, the platform executes a settlement flow:

1. Confirm winning bid $\geq$ reserve [6†L312-L317] .
2. Debit buyer wallet (or lock funds) for total amount.
3. Disburse broker commission and platform fee (configurable, e.g. 0.25–1%) automatically to respective accounts. (EATTA specifies brokers *cannot* refund any of these commissions [1†L182-L190] .)
4. Credit net sale proceeds to seller’s account.

5. Transfer eWR ownership on-chain to buyer.
6. If buyer fails to pay by Prompt Date, invoke penalties (forfeit deposit, ban) and potentially **re-auction** or relist the lot (with broker or owner permission). The system flags unsettled sales in reports.

Direct Listing (Marketplace)

TokenHarvest also supports **direct sale listings** outside auction. Producers or brokers can post a lot at a fixed price (per kg) or negotiable listing. This is useful for smaller producers or off-season offers. Core features:

- Sellers fill the same catalogue fields (Table 2) via a “New Listing” form. They set a **Buy-It-Now price** or a price range, quantity available, and optionally allow buyers to make offers.
- Buyers browse/filter listings by commodity, grade, GI, etc. They can purchase immediately (if quantity is small) or submit an offer. If it’s a partial quantity, the listing quantity is decremented upon sale (with updated eWR or separate parcel).
- Payment and settlement use the same escrow and prompt rules (buyers pay into escrow; funds release to seller when conditions met). Delivery orders are issued once payment clears.
- Unlike auctions, there are no bid increments or timing rules. However, the platform may require a price floor (reserve) just as a minimum listing price.
- The marketplace UI shows “best offer so far” if multiple buyers propose, or a countdown if seller sets an expiry date for the listing.

Comparison of Catalogue Types (Table 1): In tea auctions, **with-valuation** catalogues (common for CTC teas at Mombasa) include broker price guidance (confidential valuation) **[4†L250-L256]** ; **without-valuation** (often used for high-grown orthodox teas or direct marketing) omit any suggested price, leaving buyers to appraise. TokenHarvest supports both, toggled per listing. Regardless, all other fields (origin, grade, etc.) are mandatory.

Data Models & API Examples

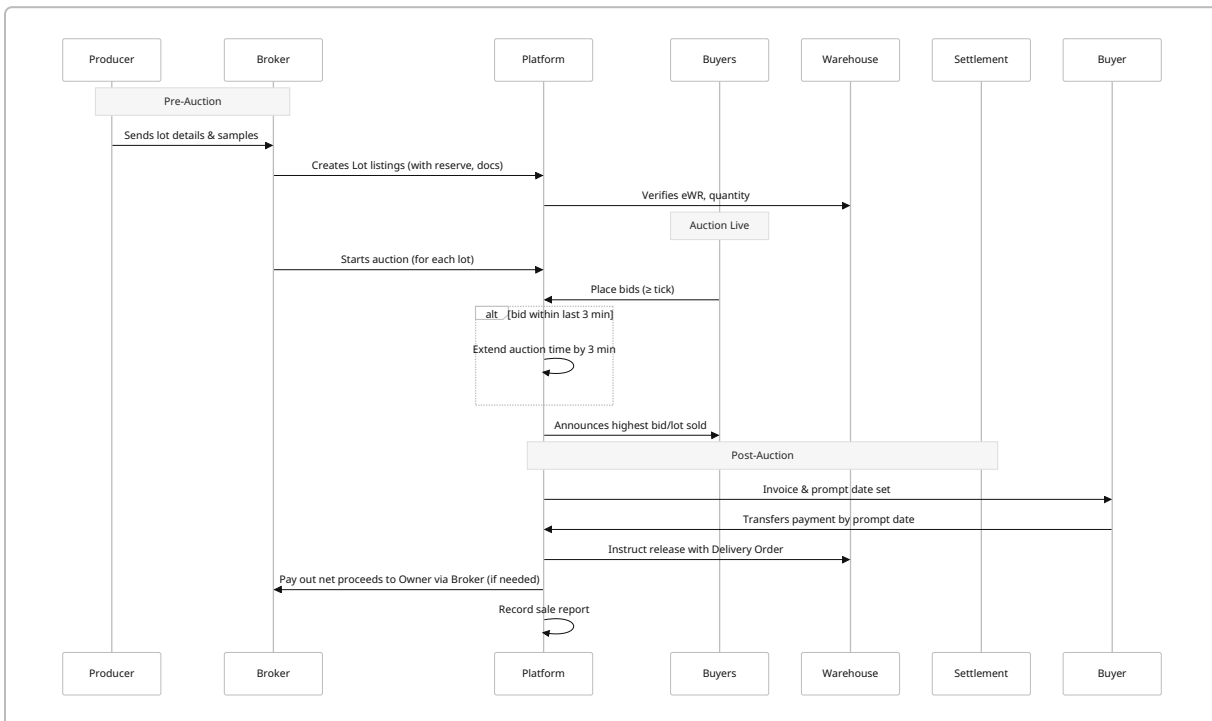
We model the core entities in the database as follows:

```
erDiagram
    USER ||--o{ BROKER : has
    USER ||--o{ BUYER : has
    USER ||--o{ WAREHOUSE : owns
    BROKER ||--o{ LOT : manages
    BUYER ||--o{ BID : makes
    WAREHOUSE ||--o{ LOT : stores
    OWNER (CommodityOwner) ||--o{ LOT : owns
    LOT ||--o{ BID : receives
    LOT ||--o{ DISPATCH : generates
    BID ||--|| USER : bidder
    BID ||--|| LOT : forLot
    SETTLEMENT ||--|| LOT : ofLot
```


SETTLEMENT ||--|| USER: buyer
 SETTLEMENT ||--|| USER: seller

- **USER:** base table (common user info, roles, wallet). Specialized by role.
- **BROKER/OWNER/WAREHOUSE:** Subtypes or role flags for USER; each with own profile (license, membership).
- **LOT:** The catalogue entry (auction lot or listing), linked to owner, broker, warehouse. Contains all fields from Table 2.
- **BID:** Records each bid (lot_id, buyer_id, amount, timestamp). High bids compute sale.
- **DISPATCH:** Records shipment events. Pre-Auction Dispatch Form entries and actual warehouse receipts.
- **SETTLEMENT:** Tracks payment for each lot (amounts, fees, status).
- **COMMISSION:** Optional separate table for broker commission calculations.

Auction Lifecycle (Sequence)



Dispatch & Documentation

Prior to each auction, producers complete a **Dispatch Form**: a list of lots (lot IDs, quantities) dispatched from factory to warehouse on a given date. This generates a Warehouse Weighment Report at arrival. The platform stores digital copies of:

- **Pre-Auction Dispatch Form:** Submitted by producer/broker, listing all lots heading to auction warehouses (dates, weights).
- **Warehouse Arrival/Weighment Report:** From warehouse, confirming net weights and any weight/grade discrepancies (endorsement to dispatch note).

- **Dispatch List (Post-Auction):** Generated by broker after sale, indicating which lots were released to which buyers, by delivery order.

All dispatch documents are stored and linked to each lot for audit and inventory tracking. Buyers cannot withdraw tea without presenting the official Delivery Order issued by their broker (after payment) [6†L381-L389] .

Settlement & Post-Auction Management

- **Invoices:** Immediately on sale, Platform issues invoices to buyer (with Tax, Currency, etc.) and account-sales to seller. Buyer must pay to TokenHarvest escrow.
- **Ownership Transfer:** Upon full payment, the eWR is transferred in the WRS registry to the buyer. Physical ownership is transferred only when the warehouse releases the lot against Delivery Order.
- **Commission Payouts:** The system splits the sale price according to commission rates. Example: If sale = KES 20,000,000; broker fee 1%; platform fee 0.5% → broker gets KES 200,000, platform KES 100,000, producer KES 19,700,000 [46†L92-L94] . Multi-tier commission (bonus for above-target prices) can be supported.
- **Penalties:** Default penalties (late payment interest, bid security forfeiture) are automatically enforced per contract terms. Default incidents are recorded. Re-auction or supplier credit arrangements may be triggered by repeated defaults.
- **Reporting:** The system generates daily/weekly/monthly reports for admins and members:
- **Sales Reports:** Summary of lots sold (value, price/kg, buyer, broker, etc.).
- **Inventory Reports:** Stocks in each warehouse by owner and collateral status.
- **Failed Transactions:** Lots unsold or defaults (with reasons).
- **Broker Performance:** Commission earned, sold volume, settlement timeliness.
- **Dispute Logs:** Any quality or quantity claims and their resolution.

Security & Compliance

- **Authentication & KYC:** Strong user authentication (2FA) and KYC verification on sign-up. Brokers and warehouses must be certified (documents on file). Buyer credit lines require AML checks.
- **Data Security:** All data (personal, financial) is encrypted at rest and in transit. Auction data is stored on secure databases with role-based access.
- **Audit Trail:** Every bid, listing change, payment and document upload is timestamped and immutable (blockchain/hashchain can be considered for ultimate audit).
- **Regulatory Compliance:** Adheres to Kenya's Warehouse Receipt System Act and regulations (WRS Council) [25†L251-L259] . Listings represent legally valid eWRs; transfers happen through the central registry. Tax/VAT invoicing complies with Kenya Revenue rules for exports.
- **Tea Standards:** Quality grades follow ICC and ISO nomenclature. E.g., "Black Tea (CTC) OP1" refers to ISO 3720 standards. Sampling and tasting protocols align with ISO 6785 (chemical) and ISO 3103 (infusion testing). Lot sampling procedures follow Tea Board guidance (4kg sample apportionment [4†L222-L227]).

UX Screens & API

Sample UI wireframes would include:

- **Catalogue Upload Screen:** Form to input all lot fields (Table 2) and attach documents.
- **Broker Dashboard:** View of owned lots (status: pending, live, sold), commission earned, quick stats.
- **Auction Live View:** Real-time updating of current lot, highest bid, bid entry interface, time remaining.
- **Dispatch Management:** Screen to generate/print Dispatch Forms and log deliveries.
- **Settlement Dashboard:** View of sales, pending payments, and auto-generated invoices.

(Images not shown here)

API Example (Lot Creation)

```
POST /api/lots
Authorization: Bearer {brokerToken}
Content-Type: application/json

{ ...lot JSON as above... }
```

Returns: `{"lotId": "LOT-20230715-001", "status": "Cataloged"}`.

API Example (Place Bid)

```
POST /api/auctions/LOT-20230715-001/bids
Authorization: Bearer {buyerToken}
Body: {"amount": 2.48}
```

Validates bid $\geq$ current+tick, returns updated high bid or error.

Assumptions & Decisions Pending

- **Currency:** All bidding uses USD/kg (like Mombasa). Local currency options may be added but need FX handling.
- **Commission Rates:** We use 1%/0.5% by default **[46†L92-L94]** , but actual structure may vary by commodity. (This is not a regulatory mandate, but market practice.)
- **Legal Venue:** Kenya/EATTA rules will govern disputes; specific legal templates (power of attorney, broker agreements) need country-specific drafting.
- **Mobile vs. Desktop:** Initial UI will target web/desktop, with responsive layout; mobile app plans TBD.
- **Tax Handling:** The platform will accommodate withholding taxes and VAT as per local laws (e.g. 16% VAT on brokerage in Kenya).

This design ensures TokenHarvest's Tea Auction Engine is an **institutional-grade commodity exchange**, combining best practices from the Mombasa and Colombo auctions with modern digital marketplace features. It streamlines pre-sale cataloging, live price discovery, and post-sale settlement while maintaining complete transparency, security, and regulatory compliance.
